

ADITYA PAL

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SUMMARY

Mechanical Engineer with hands-on experience in CAD modeling, mechanical design, prototyping, and manufacturing processes. good in SolidWorks, engineering drawings, GD&T basics, and process-oriented problem solving. Experienced in mechanical mini-projects, 3D printing workflows, and Skilled at translating user needs into practical solutions through research, prototyping, testing, and iterative improvement. Strong interest in engineering tool development, automation, productivity improvement, and industrial design applications.

EDUCATION

B.Tech - Mechanical Engineering

August 2022 -May 2026

JSS Academy of Technical Education Noida

- **Relevant coursework:** Thermodynamics, Materials science, machine design , engineering drawings. Manufacturing Processes Cad&Cam .

Senior Secondary (12th Standard — Science)

April 2020 - March 2021

Ingraham English Medium School

Technical Skills:

CAD & Design: SolidWorks (Part Modeling, Assemblies, 2D Drawings), engineering documentation , GD&T,DFM,DFA,FMEA, SLA/FDM 3D Printing Tools

Data Analysis: MS PowerPoint , MS Excel ,MS Word , MS office, AI tools

Soft skills :

- Good communication,problem-solving ,team work ,User Research, learning attitude,attention to details
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PROFESSIONL EXPERIENCE

Mechanical Desgin Intern -Ripple healthcare (R&D Department)

April 2026 - present

- Working on medical device product development and prototyping.
- Collaborated on design improvements based on testing results and user requirements.
- Supported rapid prototyping using 3D printing and iterative design validation.
- Identified design and manufacturing issues and proposed practical solutions.
- Coordinated with team members to evaluate product performance and usability.

Project 1 - Stride (Assistive device for Parkinson's patients):

- Designed enclosure and internal mounting architecture; integrated electronics and mechanical components.
- Conducted usability & ergonomics testing and iterated design into validated MVP.

Manufacturing Intern -Poly White Pvt. Ltd. (Noida)

June 2024 - July 2024

- Worked in HDPE injection molding operations for manufacturing Fischer wall plugs.
 - Learned manufacturing workflows, quality checks, and production process coordination.
 - Observed tooling, molding operations, and industrial manufacturing practices
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PROJECT EXPERIENCE

Reserch Project

Mechanical & surface characterization of woven gfrp composite reinforced with nano graphene powder

Developed GFRP composite laminates using woven glass fiber and nano-graphene reinforcement.Fabricated specimens using the hand layup technique with epoxy and polyester matrices.Investigated the effect of graphene addition on mechanical and surface properties.Compared material performance and documented experimental findings.

Push Button Enclosure Design | SolidWorks

- Designed a push-button mechanism and protective enclosure for ease of operation and durability.
- Developed 3D models considering assembly, fitment, and manufacturability requirements.
- Created technical drawings for fabrication

Achievements:

Vice President Mechanical Society: Led and organized technical events and workshops, ensuring smooth coordination and students engagement

Languages: English, Hindi